

Dr. Akash Mer, MBBS

ECFMG-certified

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RESEARCH INTEREST

How does the brain build a spatial map from the brush of air on a forearm to the position of players across a football field that is more precise than any single sense alone?

I want to unravel the process of construction of this integrated whole which is much more granular than any of its parts.

EDUCATIONAL & PROFESSIONAL BACKGROUND

GATE Scores **2026**

PAPER	ALL INDIA RANK (AIR)	TOTAL CANDIDATES
Biotechnology (BT)	97	21513
Life Sciences (XL)	287	40035

Medical Officer (MO) **2022-2023**

Sub-District Hospital, Kasa, Palghar

Bachelor of Medicine and Bachelor of Surgery (MBBS) **2011 - 2018**

Grant Government Medical College and Sir J.J. Group of Hospitals, Mumbai

Advanced Clinical Elective: Neurology **03/2018**

Hackensack Meridian Health-JFK Medical Center, Edison, New Jersey, USA

Advanced Clinical Elective: Neurology **02/2018**

Florida International University, Miami, Florida, USA

Advanced Clinical Elective: Neuroradiology **12/2017 - 01/2018**

University Hospitals, Cleveland, Ohio, USA

Pre-Medical Higher Secondary Education **2009 - 2011**

The D.G. Ruparel College of Arts, Science and Commerce, Mumbai

Core Focus: PCMB (Physics, Chemistry, Mathematics, Biology) and Information Technology

TECHNICAL SKILLS

- **EEG signal interpretation:** Long duration video-EEG review, artifact recognition, epileptiform discharge localization and characterization, video-to-EEG correlation
- **Seizure semiology:** Seizure type classification, seizure origin localization and longitudinal monitoring
- **Neuroimaging literacy:** basic physics of CT, MRI and DWI, contrast and MR sequence (T1/T2/FLAIR) mechanisms, neuroanatomical landmark recognition, lesion-symptom correlation
- **Bedside human neuroscience experience:** neurological assessment, cognitive assessment, lesion-phenotype correlation
- **Certified for Python, R (Coursera):** Data cleaning, analysis and modeling foundations
- **Knowledge management:** Obsidian + Excalidraw, visual mindscaping

INDEPENDENT PREPARATION

I built a fundamental backbone in mathematics using linear algebra, single- and multi-variable calculus, differential equations and probability and statistics. I built and iteratively revised my personal knowledge management system (PKM) into a mind-map based visual system in Obsidian/Excalidraw.

I have been tracing the field from its two edges, history and current landscape:

[The Nole Nexus](#)

CLINICAL EXPERIENCE

Medical Officer (MO)

Sub-District Hospital, Kasa

- I recognized a paroxysmal event as epileptic seizure for a patient under my care. The young patient had a history of similar events linked to her menstrual cycle but was undiagnosed. Interictal EEG later confirmed a hyperexcitability focus.
- I observed a patient with a possible head injury regain consciousness into a state of complete personality change including maternal-instinct disruption with an otherwise normal neurological exam. Imaging later confirmed the presence of an extradural hematoma.

Neurology Elective

Hackensack Meridian Health-JFK Medical Center, New Jersey

- The correlation between behavioral markers on long-duration EEG recordings and video monitoring of patients was informative of various aspects of patient's condition beyond diagnosis. In one particular patient it helped us delineate the feeling of numbness and tingling as an aura for a possible ictal event, which was confirmed by the EEG recordings.
- A hemi-spatial neglect patient responded differently based on which side she was approached from for her examination.

Neurology Elective

Florida International University, Miami

- A teenager suffering from MELAS syndrome showed the effects of living with a chronic neurological condition such as behavioral changes due to repeated stroke-like events, and lifestyle changes she could not make due to her condition.
- Several parkinsonian patients with implanted DBS devices reported the improvement that made the ordeal worth it was reduction in tremors, enabling everyday tasks.

Neuroradiology Elective

University Hospitals, Cleveland

- I learned to read a radiological image by understanding the mechanism under each modality instead of mapping radiological findings to diagnoses.

CLINICAL RESEARCH EXPERIENCE

Research Assistant, 08/2016

Department of Obstetrics and Gynecology, Grant Government Medical College, Mumbai

Supervisor: Dr. Ashok Anand

I compiled 6 months of retrospective data for the likelihood of emergency lower segment cesarean section (EmLSCS) after induction of labor with cerviprime (dinoprostone) gel. The data showed a 20% EmLSCS rate following induction, significantly below the 25% national average for tertiary government institutions ($p < 0.05$). The result was used as a baseline for department-wide protocol changes.

LANGUAGES

English (*TOEFL Score: 110/120*); Gujarati (*Native*); Hindi (*Proficient*); Marathi (*Conversational*)